

STRUCTURAL WORKS SPECIFICATION

Issued for Tender · Revision 02

Project	Construction of a Single-Family Dwelling
Project Reference	2024-0026
Location	Strasbourg, France
Client / Employer	Mr & Mrs Baumgarthen
Document Status	Issued for Tender — Rev. 02
Date	15 August 2024
Prepared by	Client Technical Advisor (on behalf of the Employer)

1. GENERAL

1.1 Scope of Works

This Specification covers the design intent, materials, workmanship and performance requirements for the substructure and superstructure concrete works of the proposed single-family dwelling for Mr & Mrs Baumgarthen. The Contractor shall read this document in conjunction with the Structural Engineer's drawings issued by the Bureau d'Études Structures (BET).

In the event of a discrepancy between this Specification and the Engineer's drawings, the Contractor shall raise a Technical Query (TQ) prior to procurement or execution. **No deviation from this Specification shall be permitted without the prior written approval of the Client's Representative.**

1.2 Codes and Standards

All works shall comply with the latest editions of the following:

- NF EN 1990 — Eurocode 0: Basis of structural design
- NF EN 1991 — Eurocode 1: Actions on structures
- NF EN 1992-1-1 — Eurocode 2: Design of concrete structures
- NF EN 1997-1 — Eurocode 7: Geotechnical design
- NF DTU 13.1 — Fondations superficielles
- NF DTU 20.1 — Ouvrages en maçonnerie
- NF DTU 21 — Exécution des ouvrages en béton
- NF DTU 23.1 — Murs en béton banché
- NF EN 206/CN — Concrete: Specification, performance, production and conformity

- Fascicule 65 — Exécution des ouvrages en béton

1.3 Definitions

- **Employer / Client:** Mr & Mrs Baumgarthen
- **Engineer / BET:** Bureau d'Études Structures appointed by the Employer
- **Architect:** Agence EK Architecte
- **Contractor:** The Main Works Contractor
- **NGF:** Nivellement Général de la France (datum)

2. SITE AND GROUND CONDITIONS

2.1 Existing Ground Levels

Based on the geotechnical survey, existing ground levels at the survey points vary between **126.30 NGF and 126.50 NGF**. The Contractor shall verify all levels on site prior to commencement of earthworks.

2.2 Finished Floor Level

The Finished Floor Level (FFL) of the Ground Floor shall be set at **127.00 NGF** as agreed with the Architect, providing a step-up of approximately 500–700 mm above existing ground level.

2.3 Ground Conditions

The bearing stratum comprises silty clays (argiles limoneuses). The Contractor's attention is drawn to the moisture sensitivity of the founding soils. Excavations shall not be left open during periods of rainfall. All foundation excavations shall be blinded with a minimum **75 mm** of C12/15 lean-mix concrete within **4 hours** of reaching formation level.

2.4 Allowable Bearing Pressures (Design Intent)

The structural design shall not exceed the following allowable bearing pressures at the underside of foundations:

Limit State	Allowable Pressure
Serviceability Limit State (SLS)	0.20 MPa
Ultimate Limit State (ULS)	0.30 MPa

3. FOUNDATIONS

3.1 Foundation Type

Foundations shall be **shallow spread footings** comprising continuous strip footings (semelles filantes) under loadbearing walls and isolated pad footings (semelles isolées) at concentrated loads, in accordance with NF DTU 13.1.

3.2 Embedment and Frost Protection

- **Minimum embedment depth (D): 1.00 m** measured from finished external ground level to underside of footing.
- **Minimum anchorage (a): 0.30 m** into undisturbed natural silty clay.
- The minimum frost protection cover of 0.90 m relative to finished external ground level shall be respected at all perimeter foundations.

3.3 Strip Footings — Minimum Dimensions

Strip footings shall be sized in accordance with the loads indicated on the Engineer's drawings but shall comply with the following Client minimum dimensions for residential construction:

Reference	Width (mm)	Depth (mm)	Notes
Standard strip (loadbearing wall)	600 min.	300 min.	Twin-bar top and bottom reinforcement
Heavy strip (under high-load wall)	800 min.	300 min.	As detailed by Engineer

3.4 Isolated (Pad) Footings — Minimum Dimensions

Reference	Plan Size (mm)	Depth (mm)
Standard isolated footing	1200 × 1200 min.	300 min.

3.5 Tie Beams (Longrines / Tirants)

Where tie beams are introduced between isolated footings or between strip footings to control differential settlement, the minimum cross-section shall be **300 × 300 mm** with 4 No. minimum HA12 longitudinal bars and HA8 links at 200 mm centres.

3.6 Concrete for Foundations

- **Concrete class: C30/37**
- **Exposure class: XC2** (foundations) / XF1 where exposed
- **Maximum water/cement ratio: 0.55**
- **Minimum cement content: 300 kg/m³**
- **Aggregate size: 20 mm nominal maximum**
- **Consistency class: S3**
- **Nominal concrete cover to reinforcement: 50 mm** to all faces in contact with ground

3.7 Reinforcement

All reinforcement shall be high-yield deformed bars **Grade B500B** in accordance with NF A 35-080-1, complying with the requirements of NF EN 10080.

4. RISING WALLS AND SUBSTRUCTURE

4.1 Crawl Space Walls (Vide Sanitaire)

Walls forming the perimeter of the crawl space shall be constructed in **reinforced concrete formwork blocks (blocs à bancher)** of minimum **200 mm** nominal thickness, filled with C25/30 concrete and reinforced with

vertical HA10 bars at 400 mm centres and horizontal HA8 bars at 400 mm centres.

4.2 Damp-Proofing

A continuous bituminous damp-proof membrane shall be installed at the top of all crawl-space walls, lapped not less than 150 mm at all joints, in accordance with NF DTU 20.1.

4.3 Crawl Space Ventilation

The Contractor shall provide ventilation openings to the crawl space at a minimum rate of **0.05 m² per 10 m²** of crawl-space floor area, evenly distributed around the perimeter.

5. SUSPENDED FLOORS

5.1 Floor System

Suspended floors over the crawl space and at first-floor level shall be constructed using a **prefabricated beam-and-block system (poutrelle-hourdis)** with a structural concrete topping, in accordance with NF DTU 23.5.

5.2 Total Floor Thickness

The nominal total floor thickness shall be **220 mm** (160 mm precast + 60 mm structural topping), confirmed by the precast manufacturer based on span and loading.

5.3 Design Loads

The floor system shall be designed for the following minimum characteristic loads in accordance with NF EN 1991-1-1:

Floor Location	Permanent Load G	Live Load Q
Ground floor over crawl space	2.00 kN/m² (200 kg/m²)	2.00 kN/m² (200 kg/m²)
First floor (residential)	2.00 kN/m² (200 kg/m²)	1.50 kN/m² (150 kg/m²)
Roof slab (terrace zones)	2.00 kN/m² (200 kg/m²)	1.50 kN/m² (150 kg/m²)
Snow load (additional)	—	0.65 kN/m²

5.4 Floor Finish Build-up

The floor finish build-up at ground floor level shall be **180 mm** thick comprising insulation, screed and floor finish. The Contractor shall coordinate slab top levels accordingly.

5.5 Frost-Resistant Concrete

At all external floor and terrace zones, the structural concrete topping shall be specified as **frost/thaw resistant** (exposure class **XF3**) or alternatively treated with a proprietary epoxy protective coating subject to the Engineer's approval.

6. SUPERSTRUCTURE

6.1 Loadbearing Walls

Internal and external loadbearing walls above ground floor shall be constructed in reinforced concrete formwork blocks or solid reinforced concrete, minimum **200 mm** thick.

6.2 Beams and Lintels

Reinforced concrete beams and lintels shall be cast in C30/37 concrete with B500B reinforcement. Minimum beam depth shall be **span / 12** unless otherwise specified by the Engineer.

6.3 Parapet Walls (Acrotères)

Perimeter parapet walls to the flat roof shall be a minimum of **400 mm** in height above finished roof level, constructed in reinforced concrete and continuously reinforced with HA10 bars at 200 mm centres each face.

6.4 Staircases

The internal staircase shall be constructed with **16 risers of 175 mm maximum** and a **going of 270 mm minimum**, in accordance with the Architect's drawings.

7. EXECUTION

7.1 Sequence

The Contractor shall not commence concreting of foundations until the formation level has been inspected and approved in writing by the Engineer or the Bureau de Contrôle.

7.2 Weather Restrictions

Concreting shall not take place when the ambient temperature is below 5 °C or above 30 °C without prior approval and protective measures.

7.3 Curing

All concrete elements shall be cured for a minimum of **7 days** by water spraying, ponding, or proprietary curing compound.

7.4 Tolerances

Construction tolerances shall comply with NF DTU 21 and shall not exceed:

- Plan position: ± 10 mm
- Verticality of walls: ± 5 mm per storey
- Floor level: ± 5 mm over any 3 m

7.5 Testing

Concrete cube tests shall be carried out at a frequency of **1 set of 3 cubes per 50 m³** placed, or per pour, whichever is more onerous.

8. INSPECTION AND CONTROL

8.1 Hold Points

The following Hold Points shall be observed by the Contractor:

1. Formation level inspection prior to blinding
2. Reinforcement inspection prior to concreting of foundations
3. Reinforcement inspection prior to concreting of rising walls
4. Pre-pour inspection of suspended floor system
5. Reinforcement inspection prior to concreting of roof parapet

8.2 Bureau de Contrôle

The Bureau de Contrôle appointed by the Employer shall have unrestricted access to the works. All non-conformances raised by the Bureau de Contrôle shall be closed out prior to handover.

9. SUBMITTALS

The Contractor shall submit the following for approval prior to commencement of the relevant works:

- Concrete mix designs for each grade specified
- Reinforcement bar bending schedules
- Manufacturer's data sheets for proprietary products (DPC, curing compound, formwork blocks, beam-and-block system)
- Method statements for foundation excavation and concreting
- Curing plan
- Inspection and Test Plan (ITP)

10. WARRANTIES

The Contractor shall provide a **decennial guarantee (garantie décennale)** covering all structural works in accordance with French construction law (Articles 1792 et seq. of the Code Civil).

— END OF STRUCTURAL SPECIFICATION —

Issued by the Client's Technical Advisor for tender purposes. The Contractor is responsible for verifying all dimensions and quantities against the Engineer's drawings and reporting any discrepancies in writing prior to procurement or construction.